

Unit = 05 (Gravitation)

Newton's Law of gravitation:-

Newton's law of gravitation states that every point mass in the universe attracts point mass with a force which is directly proportional to the product of their masses and inversely proportional to square of distance between them. The direction of force is along the line joining the particles.

Newton's law of gravitation is inverse-square law in which force is inversely proportional to the square of the distance between the bodies.

Consider two masses m_1 & m_2 are separated by distance r :



$$F \propto m_1 m_2 \quad \text{--- (1)}$$

$$F \propto \frac{1}{r^2} \quad \text{--- (2)}$$

Combining Eq (1) & (2).

$$F \propto \frac{m_1 m_2}{r^2}$$

$$F = \frac{G m_1 m_2}{r^2}$$

Where G is proportionality constant called universal gravitation constant. Its value unit is ~~6.673×10^{-11}~~ $6.673 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$. The gravitational force of attraction between around us is very small because value of G is very small. (6)

Value of 'g' on Earth Surface:-

Assume that earth is a perfect sphere of radius R_e having constant density ρ as shown in fig. Place a body of mass " m " on earth surface. Mass of earth is M_e .

Gravitational force b/w earth and body is;

$$F = \frac{G M_e m}{R_e^2} \Rightarrow (1)$$

Weight force acting on body is;

$$F = mg \Rightarrow (2)$$

Comparing eq. (1) and eq. (2)

$$mg = \frac{G M_e m}{R_e^2}$$

$$g = \frac{G M_e}{R_e^2} \Rightarrow (3)$$

This is the value of g on earth surface.

It is clear that acceleration due to gravity does not depend on mass m of the object.

Acceleration due to gravity only depends on the mass of the earth and distance between body and centre of the earth.

Value of g on moon surface is

$$g_m = \frac{G M_e}{R_m^2} \Rightarrow (4)$$

Divide eq. (3) and eq. (4)

$$g/g_m = \frac{G M_e}{R_e^2} \times \frac{R_m^2}{G M_m}$$

$$\frac{g}{g_m} = \left[\frac{M_e}{M_m} \right] \times \left[\frac{R_m}{R_e} \right]^2$$

Mass of the earth is 100 times more than mass of the moon ($M_e = 100 M_m$) and its radius is four times that of moon ($R_e = 4 R_m$)

$$\frac{g}{g_m} = 100 \times \left(\frac{1}{4} \right)^2$$

$$\frac{g}{g_m} = 6$$

$$g = 6 g_m$$

(6a)

Variation of acceleration due to gravity with altitude :-

Value of 'g' with altitude.

$$F = \frac{G M_e m}{(R_e + h)^2}, \quad g_h = \frac{G M_e}{(R_e + h)^2} \quad \text{--- (1)}$$

Value of 'g' on earth surface.

$$g = \frac{G M_e}{R_e^2} \quad \text{--- (2)}$$

Divide Eq ① by ②

$$\frac{g_h}{g} = \frac{G M_e}{(R_e + h)^2} \times \frac{R_e^2}{G M_e}$$

$$\frac{g_h}{g} = \frac{R_e^2}{(R_e + h)^2}$$

$$\frac{g_h}{g} = \frac{R_e^2}{R_e^2 \left(1 + \frac{h}{R_e}\right)^2}$$

$$g_h = g \left(1 + \frac{h}{R_e}\right)^{-2}$$

Using Binomial theorem to neglect higher order term -

$$g_h = g \left(1 + \frac{2h}{R_e}\right)$$

The space around the earth in which gravity is present is called gravitational field.

(b)
Variation of acceleration Due to gravity with Depth:-

The value of g on earth surface when earth is a perfect sphere of radius R_e

having mass M_e and uniform density " ρ " is.

$$g = \frac{GM_e}{R_e^2}$$

The density of uniform earth is given as:

$$\rho = \frac{M_e}{V_e}$$

$$M_e = \rho \times V_e$$

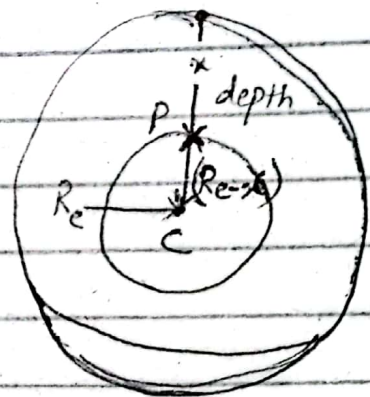
$$M_e = \rho \left(\frac{4}{3} \pi R_e^3 \right)$$

The value of g on earth surface in terms of density is -

$$g = \frac{GM_e}{R_e^2}$$

$$\cancel{M_e} = \rho$$

$$g = \frac{4}{3} \cdot \frac{G \rho \pi R_e^3}{R_e^2}$$

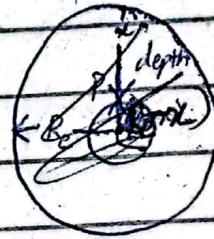


$$g = \frac{4}{3} G \rho \pi R_e$$

Consider a point P at the depth ' x ' towards centre of the earth where acceleration due to gravity is g_x .

Acceleration due to gravity at point ' P ' due to sphere of radius $(R_e - x)$ is -

$$g_x = \frac{4}{3} G \rho \pi (R_e - x)$$



$$g_x = \frac{4}{3} G \rho \pi R_e \left(1 - \frac{x}{R_e}\right)$$

$$g_x = g \left(1 - \frac{x}{R_e}\right)$$

This Eq. or formula shows that value of g decreases as we move toward earth centre.

The value of g becomes zero at the earth centre. - The body is weightless at earth centre.

At the centre of earth, $x = R_e$.

$$\begin{aligned} g_x &= g \left(1 - \frac{R_e}{R_e}\right) \\ &= g (1 - 1) \end{aligned}$$

$$\boxed{g_x = 0}$$

Density of earth :-

Volume of Spherical earth is;

$$V = \frac{4}{3} \pi R_e^3$$

$$V_e = \frac{4}{3} \pi (6400 \times 10^3)^3$$

$$V_e = 1.09 \times 10^{21} \text{ m}^3$$

Density of uniform spherical earth having mass $6 \times 10^{24} \text{ kg}$ is given as

$$\rho = \frac{M_e}{V_e}$$

$$\rho = \frac{6 \times 10^{24}}{1.09 \times 10^{21}}$$

$$\rho = 5.5 \times 10^3 \text{ kg/m}^3$$

Density of water is 10^3 kg/m^3 . Density of earth is 5.5 times more than the density water.

Kepler's Law of Area:-

Statement:-

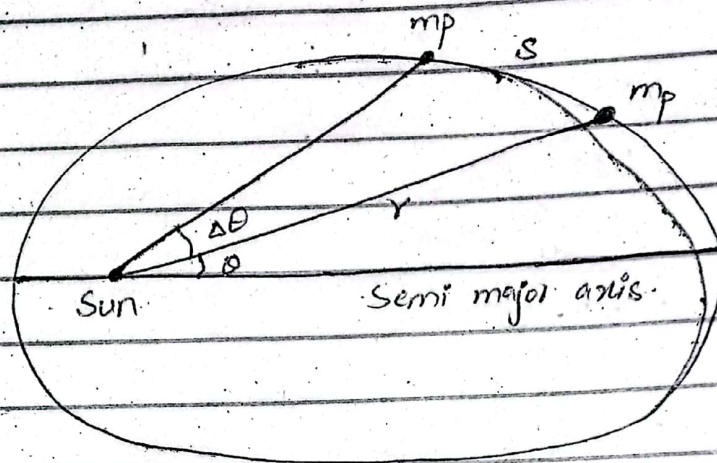
A line joining the Sun and planet sweeps out equal area in equal interval of time called Kepler's law of area.

Proof:-

Consider a planet of mass m_p is revolving around Sun which is at focus.

of an elliptical orbit as shown in fig -

The line joining the Sun and planet has length r and makes angle θ with semi-major axis.



Derivation:-

$$\text{Area of } \Delta = \frac{1}{2} b \times h$$

$$\Delta A = \frac{1}{2} r(s) \quad \text{--- (1)}$$

$$\text{Area of } \Delta = \frac{1}{2} r (r \Delta \theta) \quad \because s = r \Delta \theta$$

$$\Delta A = \frac{1}{2} r^2 \Delta \theta \quad \text{--- (2)}$$

$$\frac{\Delta A}{\Delta t} = \frac{1}{2} r^2 \frac{\Delta \theta}{\Delta t}$$

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta A}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{1}{2} r^2 \frac{\Delta \theta}{\Delta t}$$

$$\frac{dA}{dt} = \frac{1}{2} r^2 \frac{d\theta}{dt}$$

$$\frac{dA}{dt} = \frac{1}{2} r^2 \omega \quad (3)$$

$$\vec{L} = \vec{r} \times \vec{p} \quad (\text{angular momentum})$$

$$= r(mv) \sin \theta \hat{n}$$

$$= r(m_p v) \sin(90^\circ)$$

$$\text{Since } v = r\omega$$

$$\vec{L} = r(m_p r \omega)$$

$$\vec{L} = m_p r^2 \omega$$

$$\frac{L}{m_p} = r^2 \omega \quad (4)$$

Putting in eq (3)

$$\frac{dA}{dt} = \frac{L}{2m_p}$$

$$\boxed{\frac{dA}{dt} = \text{Constant}} \quad \text{Proved.}$$

Kepler's Law of Periods:~ (2)

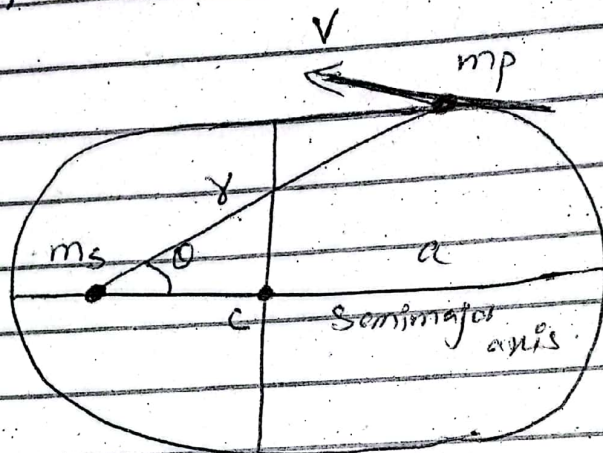
Statement :-

The Square of time period of a planet is directly proportional to cube of the mean distance between Sun and planet is called Kepler's law of periods. The ratio $\frac{T^2}{a^3}$ is constant for all

Planets orbiting the Sun -

Proof :-

Consider a planet of mass m_p is revolving around Sun having mass m_s in elliptical orbit as shown in fig. The mean ~~pos~~ distance between planet and Sun is r -



$$T^2 \propto r^3$$

$$T^2 \propto a^3$$

$$F = \frac{G m_s m_p}{r^2} \quad \text{--- (1)}$$

$$F = \frac{m_p v^2}{r} \quad \text{--- (2)}$$

Comparing (1) & (2)

$$\frac{G m_s m_p}{r^2} = \frac{m_p v^2}{r}$$

$$G_{ms} = rV^2$$

$$V = r\omega \Rightarrow V^2 = r^2\omega^2$$

$$G_{ms} = r(r^2\omega^2)$$

$$G_{ms} = r^3\omega^2$$

$$\omega^2 = \frac{G_{ms}}{r^3}$$

$$\omega = \frac{2\pi}{T}$$

$$\omega^2 = \frac{4\pi^2}{T^2} = \frac{G_{ms}}{r^3}$$

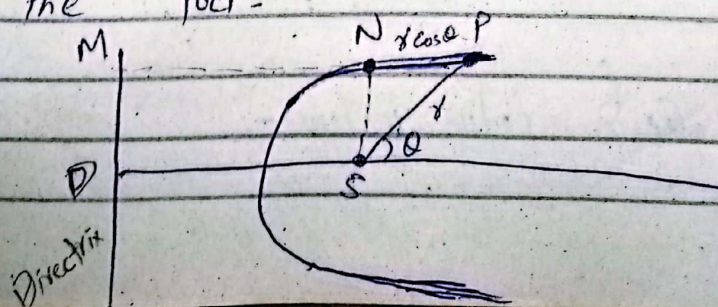
$$\frac{T^2}{r^3} = \frac{4\pi^2}{G_{ms}}$$

$$\frac{T^2}{r^3} = \text{Constant}$$

$$T^2 \propto r^3$$

Kepler's Law of orbits:-

All planets move around the Sun in elliptical orbits with the Sun at one of the foci.



Eccentricity:-

where "e" is called eccentricity of the orbit and measures the curvature of the orbit.
 e ($0 \rightarrow 1$)

$$e = \frac{r}{|PM|}$$

$$e = \frac{r}{|PN| + |PM|}$$

$$e = \frac{r}{|PN| + |Ds|}$$

$$e = \frac{r}{r \cos \theta + |Ds|}$$

$$r = e (r \cos \theta + |Ds|)$$

$$= e r \cos \theta + e |Ds|$$

$$e |Ds| = r - r e \cos \theta$$

$$e |Ds| \rightarrow L \text{ (latus rectum)}$$

$$L = r (1 - e \cos \theta)$$

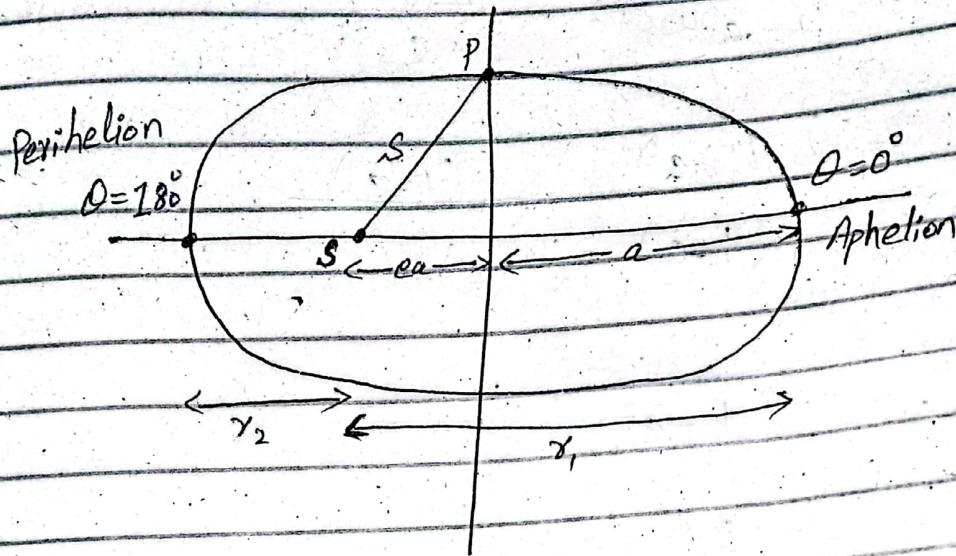
$$\boxed{r = \frac{L}{1 - e \cos \theta}}$$

(A)

$\rightarrow L$ is called latus rectum-

(3)

Sun at one Focus:-



① Aphelion:-

$$R_a = a + ae$$

$$R_a = a(1+e)$$

$$r_1 = \frac{L}{1 - e \cos Q}$$

Put $Q = 0$

$$r_1 = \frac{L}{1 - e}$$

①

② Perihelion:-

$$R_p = a - ae$$

$$R_p = a(1-e)$$

$$r_2 = \frac{L}{1 + e \cos Q}$$

Put $Q = 180^\circ$

$$r_2 = \frac{L}{1 + e}$$

②

Adding ① & ②

$$r_1 + r_2 = \frac{L}{1-e} + \frac{L}{1+e}$$
$$= \frac{L(1+e) + L(1-e)}{(1-e)(1+e)}$$

$$r_1 + r_2 = \frac{2L}{1-e^2}$$

$$2a = \frac{2L}{1-e^2}$$

$$a = \frac{L}{1-e^2}$$

Q) What is escape velocity - ? Derive formula for escape velocity using integration method - ?

Ans) **Escape Velocity:-**

The minimum velocity required for a body to leave a planet or moon is called escape velocity.

Formula of escape Velocity:-

Consider a body of mass m is placed

①

on the earth surface - Radius of earth is R_e - The body is projected with minimum velocity (V_0) to escape earth gravitational field.

Attractive gravitational force between earth and body is -

$$F = - \frac{GmM_e}{r^2} \quad \text{--- (1)}$$

Acceleration Produced in a body is -

$$F = ma$$

$$F = m \frac{dv}{dt}$$

$$F = m \frac{dr}{dr} \frac{dv}{dt}$$

$$F = m v \frac{dv}{dr} \quad \text{--- (2)}$$

Comparing Eq ① & ②

$$m v \frac{dv}{dr} = - \frac{GmM_e}{r^2}$$

$$v dv = - \frac{GM_e}{r^2} dr$$

integrate b/s.

$$\int v dv = - \int \frac{GM_e}{r^2} dr$$

$$\frac{v^2}{2} = \frac{GM_e}{r} + C \quad \text{--- (3)}$$

where C is integration Constant - ~~its~~ V_0 value.

where $r = R_e$, $V = V_0$

$$\frac{V_0^2}{2} = \frac{GM_e}{R_e} + C$$

$$C = \frac{V_0^2}{2} - \frac{GM_e}{R_e} \quad (4)$$

Put Eq (4) in Eq (3)

$$\frac{V^2}{2} = \frac{V_0^2}{2} + \frac{GM_e}{r} - \frac{GM_e}{R_e}$$

$$V^2 = V_0^2 + 2 \left[\frac{GM_e}{r} - \frac{GM_e}{R_e} \right]$$

where $r \rightarrow \infty$, $V = 0$

$$0 = V_0^2 + 2 \left[\frac{GM_e}{r} - \frac{GM_e}{R_e} \right]$$

$$V_0^2 = \frac{2GM_e}{R_e} \quad (5)$$

Value of g on earth surface is;

$$g = \frac{GM_e}{R_e^2}$$

$$2gR_e = \frac{2GM_e}{R_e} \quad (6)$$

Comparing (5) & (6)

$$V_0^2 = 2gR_e$$

$$V_0 = \sqrt{2gR_e}$$

escape velocity on earth surface is

$$V_0 = \sqrt{2 \times 9.8 \times 6.4 \times 10^6}$$

$$V_0 = 11.3 \text{ km/s}$$

escape velocity on moon surface is

$$V_m = \sqrt{2g_m R_m}$$

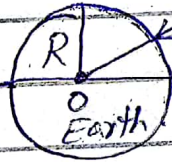
Artificial Satellite:-

The body put into orbit around earth by rocket is called artificial satellite.

The satellite having orbit close to earth is called low flying satellite.

Explanation:-

Consider a satellite having mass m revolves around earth in orbit. The gravity force provides necessary centripetal force to satellite to move in orbit.



$$F_g = F_c$$
$$mg = \frac{mv^2}{R}$$

$$v = \sqrt{gR}$$

$$v = \sqrt{9.8 \times 6.4 \times 10^6}$$

$$v = 7.9 \text{ km/s}$$

This is minimum velocity necessary to put a satellite into the orbit and is called Critical Velocity.

Time Period:-

The time period of low flying satellite is given as.

$$T = \frac{2\pi}{\omega}$$

As:

$$v = R\omega$$

$$T = \frac{2\pi R}{v}$$

$$T = \frac{2\pi (6.4 \times 10^6)}{7.9 \times 10^3}$$

$$T = 5060 \text{ sec}$$

$$T = 84 \text{ min.}$$

When a satellite is at reasonable height h above earth surface, then distance $= h + R$ is taken. The gravitational force provides centripetal force.

$$\frac{F_g}{r^2} = \frac{F_c}{r}$$
$$\frac{GmM}{r^2} = \frac{mv^2}{r}$$

Hence value of velocity is;

$$v = \sqrt{\frac{GM}{r}}$$

It shows when r is large v will be small.

Q.8) Calculate gravitational force exerted by uniform thin spherical shell on point mass which is outside shell or on surface of shell?



$$\text{Resultant vertical force} = dF_{Ay} - dF_{By}$$

$$= dF_A \sin \alpha - dF_B \sin \alpha$$

$$= \frac{G dm_A m}{r^2} \sin \alpha - \frac{G dm_B m}{r^2} \sin \alpha$$

$$\therefore [\text{use } dm_A = dm_B]$$

$$\text{Resultant vertical force} = 0$$

$$\text{Resultant horizontal force} = dF_{Ax} + dF_{Bx}$$

$$= dF_A \cos \alpha + dF_B \cos \alpha$$

(2)

$$\text{Resultant horizontal force} = \frac{Gdm_A m}{r^2} \cos \alpha + \frac{Gdm_B m}{r^2}$$

Net force is given by:

$$dF = dF_{Ax} + dF_{Bx} + dF_{Cx} + dF_{Dx} + \dots$$

$$= \frac{Gm}{r^2} (dm_A + dm_B + dm_C + \dots) \cos \alpha$$

$$= \frac{Gm}{r^2} (\text{Total mass of circular strip}) \cos \alpha$$

$$= \frac{Gm}{r^2} (\text{Density of disk} \times \text{Volume of circular strip}) \cos \alpha$$

$$= \frac{Gm}{r^2} (\rho \times \text{width} \times \text{thickness}) \cos \alpha$$

$$= \frac{Gm}{r^2} (\rho \times 2\pi R \sin \theta \times R d\theta \times t) \cos \alpha$$

$$= 2\pi \rho G m t R^2 \frac{\sin \theta \cos \alpha d\theta}{r^2}$$

$$dF = K \frac{\cos \alpha \sin \theta d\theta}{r^2} \quad \text{--- (1)}$$

Applying law of cosines:

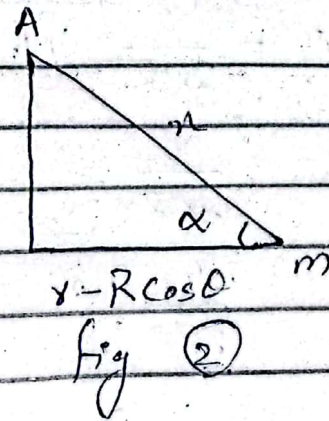
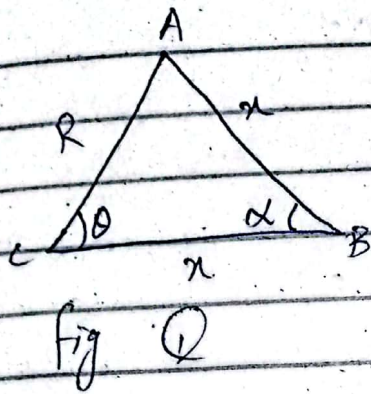
$$r^2 = x^2 + R^2 - 2xR \cos \theta \quad \text{for fig (1)}$$

$$R \cos \theta = \frac{x^2 + R^2 - r^2}{2x}$$

Diff w.r.t x

$$-R \sin \theta d\theta = \frac{0 + 0 - 2r dr}{2x}$$

(3)



$$\sin \theta d\theta = \frac{r d\alpha}{Rr}$$

From right angle triangle.

$$\cos \alpha = \frac{r - R \cos \theta}{r} \Rightarrow \frac{\text{Base}}{\text{hyp}}$$

Put value of $\cos \alpha$ and $\sin \theta d\theta$ in Eq 1

$$\begin{aligned} df &= \frac{K}{r^2} \left[\frac{r - R \cos \theta}{r} \right] \left[\frac{r d\alpha}{Rr} \right] \\ &= \frac{K}{r^2 R} [r - R \cos \theta] d\alpha \end{aligned}$$

Put value of $R \cos \theta$

$$df = \frac{K}{r^2 R} \left[r - \frac{(r^2 + R^2 - r^2)}{2r} \right] d\alpha$$

$$= \frac{K}{r^2 R} \left[\frac{2r^2 - r^2 - R^2 + r^2}{2r} \right] d\alpha$$

$$= \frac{K}{2r^2 R} \left[\frac{r^2 - R^2}{r^2} + \frac{r^2}{r^2} \right] d\alpha$$

$$df = \frac{K}{2r^2 R} \left[1 + \frac{r^2 - R^2}{r^2} \right] d\alpha$$

(4)

$$dF = \frac{K}{2x^2R} [1 + (x^2 + R^2)^{-2}] dx$$

$$F = \frac{K}{2x^2R} \int_{x=R}^{x=R} [dF] dx$$

$$= \frac{K}{2x^2R} \int_{x=R}^{x=R} [1 + (x^2 - R^2)^{-2}] dx$$

$$= \frac{K}{2x^2R} \left[x - (x^2 - R^2)^{-1} \right]_{x=R}^{x=R}$$

$$= \frac{K}{2x^2R} [(x+R) - (x-R)] \left\{ (x+R)^{-1} - (x-R)^{-1} \right\}$$

$$= \frac{K}{2x^2R} [2R - (x+R)(x-R)] \left\{ (x+R)^{-1} - (x-R)^{-1} \right\}$$

$$= \frac{K}{2x^2R} [2R - (x+R)(x-R) + (x+R)(x-R)]$$

$$= \frac{K}{2x^2R} [2R - (x-R) + (x+R)]$$

$$= \frac{K}{2x^2R} [2R - x + R + x + R]$$

$$= \frac{K}{2x^2R} (4R) \Rightarrow \frac{24KR}{2x^2R}$$

$$F = \frac{2K}{x^2}$$

put value of $K = 2\pi G_m \rho L R^2$

$$F = 2 \times 2\pi G_m \rho L R^2 \cdot \frac{1}{x^2}$$

$$F = G_m \cdot \frac{4\pi R^3 \rho L}{x^2}$$

(5)

$$F = \frac{G_m (\text{Surface area} \times \text{density} \times \text{thickness})}{y^2}$$

$$= \frac{G_m (\text{Volume} \times \text{density})}{y^2}$$

$$F = \frac{G_m M}{y^2}$$

Put $y=R$ when mass 'm' lies at surface of the shell.

$$F = \frac{G_m M}{R^2}$$

Gravitational force of shell on point mass lies inside:-

$$F = \frac{K}{2y^2R} \left[x - (y^2 - R^2)x^{-1} \right]_{R-y}^{R+y}$$

$$F = \frac{K}{2y^2R} \left[(R+y) - (R-y) - (y^2 - R^2) \left\{ \frac{1}{(R+y)} - \frac{1}{(R-y)} \right\} \right]$$

$$F = \frac{K}{2y^2R} \left[(R+y) - \frac{(y^2 - R^2)}{(R+y)} - (R-y) - \frac{y^2 - R^2}{(R-y)} \right]$$

$$F = \frac{K}{2y^2R} \left[(R+y) - \frac{(y-R)(y+R)}{(R+y)} - (R-y) - \frac{(y-R)(y+R)}{(R-y)} \right]$$

$$F = \frac{K}{2y^2R} \left[(R+y) - (y-R) - (R-y) - \frac{(R-y)(R+y)}{(R-y)} \right]$$

$$F = \frac{K}{2y^2R} \left[(R+y) - (y-R) - (R-y) - (y+R) \right]$$

$$F = \frac{K}{2y^2R} \left[(R+y) + (R-y) - (R-y) - (y+R) \right]$$

$$F = 0$$